NISHI AGARWAL

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Education

National Institute of Technology, Bhopal 2023 – Present
B.Tech, Electronics and Communication Engineering *CGPA: 8.03*

D.A.V. Academy School, UP 2023
Board of Intermediate Education, Uttar Pradesh *Percentage: 93.6%*

Lord Buddha Public School, UP 2021
Board of Secondary Education, Uttar Pradesh *Percentage: 97.6%*

Technical Skills

Programming: Python, C++, SQL
Backend & Databases: FastAPI, PostgreSQL, Redis, MySQL, MongoDB, Docker
AI / ML: FAISS, sentence-transformers, PyMuPDF, Llama 3, NumPy, Retrieval-Augmented Generation (RAG)
Testing & Tools: pytest, Locust, Git, GitHub
System Design: Low-Level Design (OOP, SOLID, design patterns), REST API design, caching, rate limiting, indexing
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems
Core Competencies: Leadership & Team Management, Effective Communication, Problem-Solving, Adaptability, Teamwork & Collaboration

Projects

Snip: High-Throughput URL Shortener with Redis Caching | *Python, FastAPI, PostgreSQL, Redis, Docker*

- Built a REST URL-shortening service with base62-encoded IDs (7 characters, 3.5T+ unique codes), custom aliases, link expiry, and SHA-256 URL deduplication.
- Cut p95 redirect latency from **38 ms to 12 ms** using a **Redis** cache-aside layer with negative caching, reaching a **94% cache hit rate** across 50,000 requests in a Locust load test.
- Moved click tracking off the request path with Redis counters batch-flushed to PostgreSQL every 10 s; added per-IP rate limiting (10 req/min) and 24 **pytest** integration tests.

SpecSeek: RAG Assistant for Electronic Datasheets | *Python, FastAPI, FAISS, sentence-transformers, Llama 3*

- Built a retrieval-augmented QA system answering questions over component datasheets with page-level citations, using PyMuPDF extraction, overlapping chunking, and **FAISS** cosine-similarity search.
- Raised retrieval hit rate@4 from **68% to 84%** on a 25-question evaluation set by benchmarking chunk sizes of 400, 800 and 1200 characters.
- Curbed hallucinations with a similarity threshold that refuses out-of-scope questions before calling the LLM (4/5 correct refusals); hash-based embedding cache makes re-uploads skip re-indexing entirely.

Achievements

- Solved **500+ DSA problems** across **LeetCode** and **CodeChef**.
- Secured **All India Rank 1657** among 10,000+ participants in the Goldman Sachs India Hackathon (GSIH) 2026.
- Secured **All India Rank 23340** in JEE Mains 2023 among **1M+** candidates.

Positions of Responsibility

Event Management and Competitions Head March 2024 – Present
Tooryanaad

- Organized a three-day cultural fest featuring 13 diverse competitions, managed 2000+ registrations and 800+ entries nationwide, coordinated with judges, and executed extensive online and offline promotions.

Technical Head April 2024 – Present
Kairos

- Founding member responsible for managing and overseeing the society's organizational and technical aspects.
- Organized the Internal Hackathon for **Smart India Hackathon** at our college with 100+ team registrations and 600+ participants, collaborated with 40 jury members (professors and industry experts), ensured smooth management and data accuracy.